

MScProject: Evidence-Grounded Table-to-Text Generation

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Purpose

This software converts structured tabular data into natural-language reports while reducing unsupported factual claims. It was developed for an MSc project investigating whether an evidence-grounded multi-agent workflow can improve table-to-text generation over a direct single-model baseline. The system does not claim to eliminate hallucinations; it makes evidence provenance, claim verification, sentence support, and release decisions explicit and auditable.

Program Operation

The input may be CSV, JSON, spreadsheet, Parquet, or Arrow data together with a natural-language reporting request. Deterministic code loads and profiles the data, interprets supported analytical capabilities, executes analyses, and records evidence identifiers. Six semantic roles then cooperate: Data Understanding interprets structure and field meaning; the Orchestrator freezes a capability-aware plan; the Evidence Analyst proposes grounded claims and bounded insights; the Claim Verifier checks support and permissions; the Writer realises verified content under a report contract; and the Factual Auditor checks factual support and task fulfilment. Deterministic validators retain control of provenance, numerical permissions, support maps, and release gates.

The workflow supports dataset overviews, data-science reports, focused table descriptions, attribute and triple verbalisation, and event reports. It can run with configured LLM services or in a deterministic fallback mode that requires no API key. Each run creates a timestamped artifact directory containing the interpreted structure, execution plan, evidence and fact ledgers, writer output, audit records, final report, and machine-readable result.

Interfaces and Evaluation

The installable wheel exposes `table2text` for generation and `table2text-evaluate` for dataset preparation, report generation, automatic metrics, diagnostics, human-study export, and aggregation. A Python API and research notebooks are also supplied. Automated tests cover loading, capability selection, semantic event handling, evidence validation, writing, auditing, and reference-based evaluation.

The included protected evaluation evidence contains 25 unseen examples across five heterogeneous datasets, paired Full System and Baseline outputs, reference-alignment and source-grounded metrics, model-input audits, and structured judge results. Held-out references are isolated from generation.

Deliverables

The submission contains maintained source and test programs, configuration, notebooks, curated data/results, full program listings, installation and build instructions, exact dependency versions, a deterministic demonstration, and a platform-independent Python wheel. Credentials, downloaded caches, local model weights, virtual environments, and transient development runs are intentionally excluded.